

The Ultimate Web Development Handbook



Introduction to CSS

CSS (Cascading Style Sheets) is used to style and layout web pages — including colors, fonts, spacing, and positioning of elements. While HTML gives structure to a web page, CSS makes it look beautiful and usable.

Why CSS?

Without CSS, all websites would look plain, like unstyled documents. CSS helps you:

- Change colors and fonts
- Add spacing and layout
- Make responsive designs for mobile
- Animate and transition between states
- Separation of concerns: HTML for structure, CSS for style

How CSS Works with HTML

CSS can be applied to HTML in **three main ways**:

1. Inline CSS

CSS written inside an HTML tag using the `style` attribute.

```
<p style="color: blue; font-size: 18px;">This is a blue paragraph.</p>
```

2. Internal CSS

CSS written inside a `<style>` tag within the `<head>` section of the HTML.

```
<!DOCTYPE html>
<html>
  <head>
    <style>
      p {
        color: green;
        font-weight: bold;
      }
    </style>
  </head>
  <body>
    <p>This is a green bold paragraph.</p>
  </body>
</html>
```

3. External CSS (Best Practice)

CSS written in a separate file and linked to the HTML file. This is the **most recommended** method for real-world projects.

style.css

```
h1 {
  color: darkred;
  text-align: center;
}
```

index.html

```
<!DOCTYPE html>
<html>
  <head>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <h1>Welcome to CSS</h1>
```

```
</body>
</html>
```

The “Cascading” in CSS

If there are multiple rules targeting the same element, CSS uses the **cascade** to decide which rule to apply. This depends on:

- **Specificity** (How specific the selector is)
- **Order of appearance**
- **Importance** (`!important`)

Example:

```
<p style="color: red;">This will be red because inline CSS wins.</p>
```

Anatomy of a CSS Rule

```
selector {
  property: value;
}
```

Example:

```
p {
  color: black;
  font-size: 16px;
}
```

- `p` → Selector (targets all `<p>` elements)
- `color` , `font-size` → Properties
- `black` , `16px` → Values

What You'll Learn in CSS

As we move forward, you'll learn how to:

- Style text, backgrounds, and borders
- Control layout with Flexbox and Grid
- Make your website responsive and mobile-friendly
- Animate elements and transitions
- Write modern, maintainable CSS

CSS Syntax and Selectors

To apply styles to HTML elements, you need to understand the basic **syntax of CSS** and how to **select elements** on the page.

CSS Syntax

Every CSS rule consists of a **selector** and a **declaration block**.

```
selector {  
  property: value;  
}
```

Example:

```
h1 {  
  color: navy;  
  font-size: 32px;  
}
```

- `h1` is the **selector**
 - `color` and `font-size` are **properties**
 - `navy` and `32px` are the **values**
 - The curly braces `{ }` contain the **declaration block**
 - Each declaration ends with a semicolon `;`
-

Types of Selectors

1. Element Selector

Selects all elements of a specific type.

```
p {  
  color: gray;  
}
```

This targets all `<p>` elements.

2. Class Selector

Selects elements with a specific class.

HTML:

```
<p class="highlight">This is important.</p>
```

CSS:

```
.highlight {  
  background-color: yellow;  
}
```

Use a period `.` before the class name.

3. ID Selector

Selects a single element with a unique ID.

HTML:

```
<h1 id="main-heading">Welcome</h1>
```

CSS:

```
#main-heading {  
  font-family: Arial, sans-serif;  
}
```

Use a hash # before the ID name.

4. Universal Selector

Applies styles to all elements on the page.

```
* {  
  margin: 0;  
  padding: 0;  
}
```

This is commonly used for resetting default styles.

5. Grouping Selectors

Apply the same styles to multiple selectors at once.

```
h1, h2, h3 {  
  color: darkblue;  
}
```

This avoids repetition.

6. Descendant Selector

Targets elements nested inside other elements.

HTML:

```
<div>
  <p>This is a paragraph inside a div.</p>
</div>
```

CSS:

```
div p {
  font-style: italic;
}
```

Only `<p>` tags inside `<div>` will be affected.

7. Combining Class and Element Selectors

You can be more specific by combining them.

```
p.note {
  color: teal;
}
```

This targets only `<p>` elements with the class `note`.

Summary

- CSS selectors help you choose **which** HTML elements to style.
- Use `.` for classes, `#` for IDs, and tag names for element selectors.
- Combine and group selectors for powerful control.